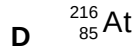
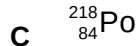
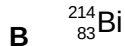
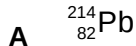


29 Radon-222, $^{222}_{86}\text{Rn}$ decays to Lead-210, $^{210}_{82}\text{Pb}$ via a series of three alpha and two beta decays through a series of intermediate nuclides. Which of the following cannot be one of the intermediate nuclides produced?



Ans: D

Alpha decays cause nucleon number to decrease by 4 each time.

Beta decays do not affect nucleon number.

Possible nuclides can only have nucleon numbers of 218, 214, 210.

(Sufficient to only consider nucleon number in this case.)

30 An experiment is carried out in which the count rate is measured at a fixed distance from a